

From Reflection Positivity to a Uniform Reconstructed Gap: A Machine-Checked Transfer Operator for Coupled $\mathbb{Z}_2$ Slices

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Abstract

We close a compositional gap between two previously machine-checked finite-volume constructions. The first reconstructs a transfer operator from the positive site form forced by reflection positivity. The second proves a volume-uniform Dobrushin spectral gap for a concrete tilted kernel. We define the square-root dressing of real boundary vectors, prove that it is an isometry for the reconstructed site form, identify the forced transfer operator with the symmetric weighted kernel, transport this identity through all finite iterates and Perron normalisation, and finally compose it with the existing Dobrushin theorem. Under

$$2 \tanh |\beta| + 2 \tanh |\gamma| \leq \alpha < 1,$$

one positive exponent works for every spatial extent. The six interface theorems are checked by Lean 4 against Mathlib. The contribution is the exact machine-checked composition and its binder order, not a new analytic estimate.

1 Scope and claim discipline

Osterwalder–Schrader (OS) reflection positivity turns Euclidean data into a physical Hilbert-space structure and a time-translation operator [1, 2, 3]. In finite volume the relevant algebra is elementary, but a formal development still has to identify which positive form is physical, prove that the operator is forced by that form, and connect its normalised spectrum to a mixing estimate whose constants do not deteriorate with the spatial size.

This paper proves precisely that interface for the coupled $\mathbb{Z}_2$ slice model already developed in the repository. It does *not* claim a new Dobrushin estimate, a continuum limit, a Wightman reconstruction, or the first formal treatment of reflection positivity. In particular, recent Lean work verifies Euclidean axioms for the free bosonic field [6]. Our narrower contribution is a concrete reconstructed finite-volume transfer operator whose normalised fluctuation operator inherits a single, explicit, volume-uniform gap exponent.

2 Finite slice data

For $L \in \mathbb{N}$, let

$$\Sigma_L = (\text{Fin } L \rightarrow \text{Fin } 2)$$

be the finite set of spin configurations on a spatial slice. Fix a positive weight $w : \Sigma_L \rightarrow \mathbb{R}_{>0}$ and the real symmetric spatial kernel $K_\beta(\sigma, \tau)$. The reconstructed site form on complex boundary vectors is

$$\langle u, v \rangle_{\text{site}, w} = \sum_{\sigma \in \Sigma_L} \overline{u(\sigma)} v(\sigma) w(\sigma)^{-1}. \quad (1)$$

The transfer operator forced by the site–bond identity is

$$(T_{w,\beta}v)(\sigma) = w(\sigma) \sum_{\tau \in \Sigma_L} K_\beta(\sigma, \tau) v(\tau). \quad (2)$$

The symmetric weighted kernel is

$$S_{w,\beta}(\sigma, \tau) = \sqrt{w(\sigma)} K_\beta(\sigma, \tau) \sqrt{w(\tau)}. \quad (3)$$

All three objects precede the present module. The missing step was to state and verify the exact real-to-complex dressing through which the Dobrushin lane can be read on the reconstructed operator.

Define

$$(Q_w u)(\sigma) = \sqrt{w(\sigma)} u(\sigma), \quad u : \Sigma_L \rightarrow \mathbb{R}, \quad (4)$$

where the result is regarded as a complex boundary vector. Since $w > 0$, this is the real restriction of the already available complex linear equivalence given by multiplication by $\sqrt{w}$.

3 The six machine-checked interfaces

The new file `YangMills/OS/OSReconstructionUniform.lean` contains one definition and the following six theorem declarations.

Lean declaration	Mathematical content
<code>siteForm_qEmbed</code>	Q_w is an isometry from the real Euclidean pairing to (1).
<code>transferOp_qEmbed</code>	$T_{w,\beta} Q_w = Q_w S_{w,\beta}$ on real boundary vectors.
<code>transferOp_iterate_qEmbed</code>	The same conjugation holds for every finite iterate.
<code>transferOp_qEmbed_tilt</code>	Perron normalisation corresponds to the Dobrushin tilt in one step.
<code>transferOp_qEmbed_tilt_iterate</code>	The normalised conjugation holds for every finite iterate.
<code>os_reconstruction_uniform_gap</code>	One positive exponent and the full reconstructed identity hold for all L .

The first identity is exact:

Proposition 3.1 (Dressing isometry). *For $w(\sigma) > 0$ and real u, v ,*

$$\langle Q_w u, Q_w v \rangle_{\text{site}, w} = \sum_{\sigma \in \Sigma_L} u(\sigma) v(\sigma).$$

Indeed, each summand contains $\sqrt{w(\sigma)}^2 / w(\sigma) = 1$. In Lean the positivity hypothesis supplies both the square-root identity and the nonzero denominator.

The second identity is the actual bridge:

Proposition 3.2 (Forced-operator conjugation). *For every real boundary vector u ,*

$$T_{w,\beta}(Q_w u) = Q_w(S_{w,\beta} u). \quad (5)$$

Consequently, for every $n \in \mathbb{N}$,

$$T_{w,\beta}^n(Q_w u) = Q_w(S_{w,\beta}^n u).$$

The one-step proof reuses the previously checked complex conjugation theorem; the iterate theorem is induction on n . Thus the new file introduces no parallel definition of the physical transfer operator.

4 Perron tilt and the uniform endpoint

Let $\lambda \in \mathbb{R}$ be the Perron eigenvalue and define the real tilt matrix

$$P_{w,\beta,\lambda}(\sigma, \tau) = \lambda^{-1} S_{w,\beta}(\sigma, \tau). \quad (6)$$

Scalar multiplication on the complex function space and matrix action on the real function space then satisfy

$$\lambda^{-1} T_{w,\beta}(Q_w u) = Q_w(P_{w,\beta,\lambda} u), \quad (7)$$

and the same equation holds after every finite iterate. Formally, (7) is division-free as a statement about the inverse scalar: the proof merely distributes the same inverse through a finite sum. The final endpoint supplies $\lambda > 0$.

For the coupled-slice weight $w = \text{sliceW}(\gamma, L)$, the existing Dobrushin endpoint produces $m > 0$ independent of L , a positive Perron value λ_L , and a positive vector Ω_L . The new composition is the following statement. The library parameter L indexes the configuration space Σ_{L+1} in this endpoint; we preserve that convention instead of silently rebasing the formal binder.

Theorem 4.1 (Volume-uniform reconstructed gap). *Let $0 < \alpha < 1$ and suppose*

$$2 \tanh |\beta| + 2 \tanh |\gamma| \leq \alpha.$$

Then there is $m > 0$ such that for every $L \in \mathbb{N}$ there are $\lambda_L > 0$ and $\Omega_L : \Sigma_{L+1} \rightarrow \mathbb{R}_{>0}$ with

$$\sum_{\tau} P_L(\sigma, \tau) \Omega_L(\tau) = \Omega_L(\sigma),$$

$$\|\text{projectedTransfer}(\text{opOf } P_L, \text{vacOf } \Omega_L)\| \leq e^{-m},$$

and, for every real boundary vector u ,

$$\lambda_L^{-1} T_{\text{sliceW}(\gamma, L), \beta}(Q_{\text{sliceW}(\gamma, L)} u) = Q_{\text{sliceW}(\gamma, L)}(P_L u).$$

Here $P_L = P_{\text{sliceW}(\gamma, L), \beta, \lambda_L}$.

The quantifier order is essential: m is chosen before L . This is the precise sense in which the reconstructed gap is volume-uniform.

5 Relation to prior work

The classical reconstruction and reflection-positivity literature supplies the conceptual framework [1, 2, 3, 4]. Dobrushin’s uniqueness theory supplies the high-temperature contraction mechanism [5]. Neither fact is claimed as new here.

Work	Machine-checked content	Difference from the present endpoint
Douglas et al. [6]	Euclidean/Glimm–Jaffe axioms for a free bosonic field	Axiom verification, not a concrete reconstructed transfer operator with a measured uniform gap.
Classical OS papers [1, 2]	Analytic reconstruction theorem	Not proof-assistant checked and not specialised to this finite coupled-slice kernel.
Osterwalder–Seiler [3]	Lattice gauge reconstruction and positivity	Classical framework; no Lean object or repository-wide axiom report.
This paper	Six Lean interfaces joining a forced reconstructed operator to an existing Dobrushin gap	Finite $\mathbb{Z}_2$ slices only; no continuum or Wightman limit.

A survey of public formal artifacts performed on 4 August 2026 found no prior artifact with this exact conjunction: a concrete OS-reconstructed transfer operator, an explicit Dobrushin window, and a gap exponent quantified before the spatial volume. Because repository-level and unindexed work can be missed, we use this only as a qualified comparison, never as an unqualified “first OS reconstruction” claim.

6 Verification and reproducibility

The authoritative source is the Lean file, not this prose. Verification used a Colab Pro+ CPU, high-RAM runtime with GPU disabled. The clean target build completed 8191 jobs with exit code zero between 16:25:28Z and 16:25:39Z on 4 August 2026. The source contained exactly six theorem declarations. Its SHA-256 digest was

c474ba533c25c46f920ce70284980998bcdadef8c246ec0e38a1240102b3e14,

and the materialised `.olean` digest was

0751d3b593c3f845175cf05e1e863ced46343748796885a76705046d8c3cad0e.

The repository-wide oracle count for the frozen object is 3,046 reports: 3,019 using only standard axioms and 27 axiom-free; zero nonstandard axioms and zero `sorry`. A separate manifest records the branch SHA, byte counts, line-ending hashes, capture method, full command transcript, and the attacks attempted. Successful compilation is evidence of elaboration, not a terminal audit verdict; the frozen object is intended for independent review.

7 Limitations

The result is finite-volume and model-specific. It does not construct a continuum field theory, prove the full OS reconstruction theorem in generality, or establish a Yang–Mills mass gap. The physical space used here is the finite reconstructed site space already identified by the reflection-positive slice construction. The analytic small-coupling window and the operator-norm estimate are inherited without alteration from the Dobrushin lane. Finally, the formalisation does not turn a bibliographic survey into a mathematical priority theorem; comparison language remains deliberately qualified.

8 Conclusion

The square-root dressing Q_w is the missing typed interface between the reflection-positive reconstruction and the symmetric kernel used by the Dobrushin argument. Once that interface is stated, the one-step equality, its iterates, the Perron tilt, and the volume-uniform endpoint form a short but fully checkable chain. The resulting theorem says exactly what is uniform, exactly which reconstructed operator carries the gap, and exactly which parts are inherited.

References

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